

Final Project

Nguyen Van Thanh

ID: 20242084

1 Problem 1: Cart-Pole with Actuator Dynamics: DDP and iLQR

Prerequisite: cart-Pole Dynamics

The dynamic equation of cart-pole system is as follows:

$$\ddot{x} = \frac{\ell m_p \sin(\theta) \dot{\theta}^2 + u + m_p g \cos(\theta) \sin(\theta)}{m_c + m_p (1 - \cos^2(\theta))} \quad (1)$$

$$\ddot{\theta} = -\frac{\ell m_p \cos(\theta) \sin(\theta) \dot{\theta}^2 + u \cos(\theta) + (m_c + m_p) g \sin(\theta)}{\ell m_c + \ell m_p (1 - \cos^2(\theta))} \quad (2)$$

(a)

Denote